

GEORGIA INSTITUTE OF TECHNOLOGY
COE3001-M MECHANICS OF DEFORMABLE BODIES

Final exam - 2026 Spring

May. 1, 8:00-10:50 am

Name: _____ GTID: _____

Please read the following instructions

- ✧ You have **170** minutes to complete this exam.
- ✧ You may use a single, double-sided, letter sized formula sheet.
- ✧ This question booklet contains **6** questions, **13** pages (including the cover) for a total of **40 points + 3 bonus**. Check to see if any pages are missing.
- ✧ The maximum achievable score is **40**, including any bonus points.
- ✧ DO NOT scribble or do rough work or make any stray marks on it. Use separate sheet for rough work.
- ✧ All the questions are compulsory, and all the notations have their usual meaning.
- ✧ Show all necessary steps for full credit. Clearly indicate your final answers.
- ✧ Partial credit will be given for showing your work

Section 1 – Physical Intuition and Conceptual Reasoning (15 pts)

Answer all questions. Unless explicitly requested, lengthy calculations are not required. Credit will be based on the correctness of the answer and the quality of the mechanical reasoning. Where appropriate, support your answer with a brief sketch, limiting case, or physical argument.

Problem 1 – Foundational questions

1.1 Bending stiffness and cross-sectional orientation (0.5+1+1.5 pts)

A prismatic beam with a rectangular cross section of width b and height h ($h > b$) is subjected to transverse vertical force in two different orientations:

- Case A: the dimension h is vertical
- Case B: the beam is rotated by 90° , so the dimension b is vertical

Assume the material, beam length, and force are the same in both cases.

- (a) Which case has the smaller deflection? Case A or Case B
- (b) Which case develops the smaller bending stress at the outer surface under the same bending moment?
Case A or Case B
- (c) Explain physically why changing the orientation has such a large effect.

1.2 Stress transformation under pure shear (0.5+0.5+1 pts)

Under plane stress assumption, a material element experiences a biaxial hydrostatic stress state:

$$\sigma_y = \sigma_x \neq 0, \quad \tau_{xy} = 0$$

- (a) Are the principal stresses zero or nonzero?
- (b) On planes rotated 45° from the original faces, does the maximum shear stress appear?
- (c) Describe all the possible stress states for this material element.

Problem 2 – Counterintuitive mechanics and higher-insight questions

These questions are designed to test whether you can distinguish between **stress, strain, displacement, stiffness, strength, and stability**. Many of the statements below sound plausible at first glance. Read carefully.

2.1 True / False without Justification (6*0.5 pts)

For each statement, indicate whether it is True or False, **no justification needed or graded**.

- (a) A point in a structure can have zero displacement while the stress at that point is nonzero.
- (b) A member with zero total axial strain must have zero axial stress.
- (c) Increasing the cross-sectional area of a column always greatly increases its buckling load.
- (d) A cross section where the internal force is zero must be stress-free.
- (e) A member under an external force parallel to the axial direction only deform axially.
- (f) A point under uniaxial normal stress can undergo shear failure.

2.1 True / False with Justification (7*1 pts)

For each statement, indicate whether it is True or False, and justify your answer briefly but precisely. **A correct choice without justification will receive limited credit.**

- (a) If a structure is made stiffer (increase modulus), the stress everywhere in the structure must decrease.
- (b) A column with a larger area moment of inertia about the bending axis is generally harder to buckle.
- (c) Two beams made of the same material, with the same length and the same volume, must have similar bending stiffness.

- (d) At a point on the neutral axis of a beam, the material is necessarily unstressed.
- (e) An eccentrically applied axial load is mechanically equivalent to an axial load plus a bending moment.
- (f) A member can develop internal stress even when no external mechanical force is applied.
- (g) A fixed-fixed column is generally more resistant to buckling than a pinned-pinned column of the same length, section shape and material.

Section 2 – Computational problems

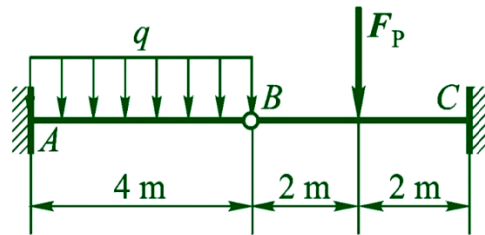
Problem 3 (6 pts)

Beams AB and BC are pin-connected at B, with ends A and C fixed. Both beams have the same flexural rigidity EI . The applied concentrated force is $F_P = 40$ kN, and the uniformly distributed load is $q = 20$ kN/m. Ignore the horizontal reactions.

(a) calculates the vertical reaction forces and reaction bending moments on A and C ends.

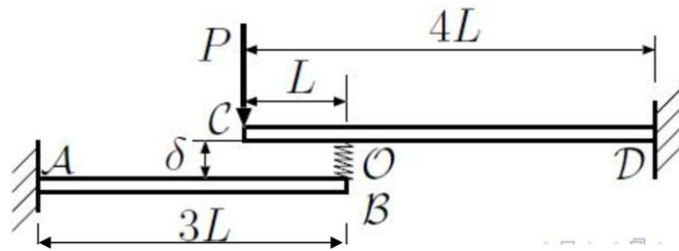
(b) Draw the SFD and BMD for the beam.

(Hint: The structure is statically indeterminate.)



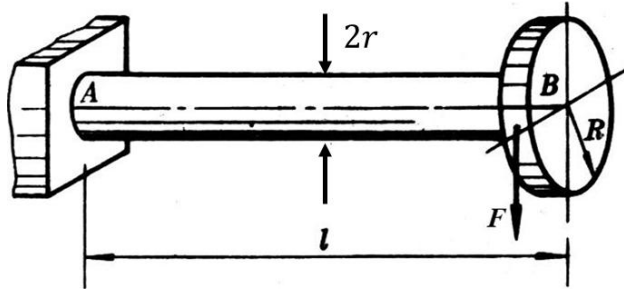
Problem 4 (7 points + 3 bonus)

As shown below, two prismatic cantilever beams AB and CD, both having Young's modulus E and moment of inertia I , are fixed at A and D, respectively. On beam AB, a spring with a constant k is fixed to the free end B. Initially with no system loading, the spring has a length of δ ($\delta \ll L$) and its tip touches the beam CD at O with no force. Now a vertical force P is applied downward at C on beam CD. Determine the magnitude of P , such that at equilibrium, beam CD begins to make contact with beam AB by its free end C. Assume all contacts occurred are smooth and the horizontal displacement can be neglected. Also, for simplicity, take $EI = 9kL^3$ and express the result in terms of k and δ .



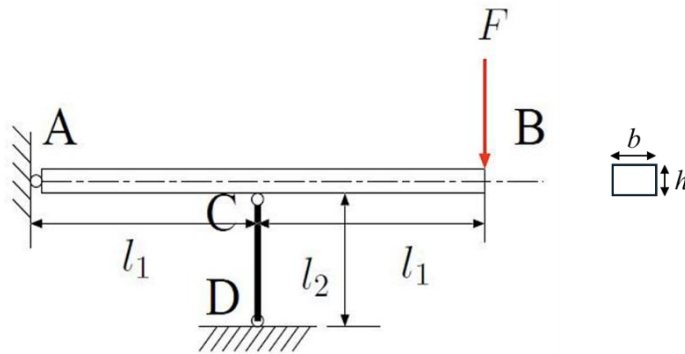
Problem 5 (6 points)

As shown below, for a solid circular shaft AB, one end A is fixed, and the free end B is assembled with a wheel. The external load F is applied on the edge of wheel. The length of the bar AB is l , the radius of the wheel is R , and the radius of the shaft is r . Determine the most dangerous point(s) according to the Maximum shear stress theory (Hint: $\sigma_{r3} = \sigma_1 - \sigma_3$). What is the maximum principal stress at the dangerous point(s)?

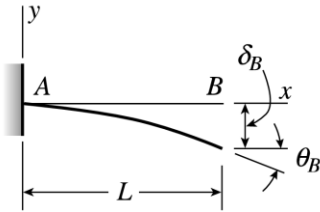


Problem 6 (6 points)

A horizontal beam with rectangular cross-section AB is pinned to a vertical wall at point A and loaded by a concentrated downward force F at its right end B. The beam is additionally supported at point C by a vertical solid circular rod CD, which is pinned at both end C and D, as shown in the figure. Given $l_1 = 1.25$ m, $l_2 = 0.55$ m, $b = 15$ cm, $h = 10$ cm. The allowable stress in AB is $[\sigma] = 180$ MPa. For the rod CD, the diameter $d = 25$ mm, Young's modulus $E = 200$ GPa, the proportional limit $\sigma_{pl} = 200$ MPa. The **linear relation of inelastic buckling** has coefficient $a = 310$ MPa and $b = 1.14$ MPa ($\sigma_{cr} = a - b\lambda$, λ is the slenderness ratio). The safety factor of stability is $n_{st} = 2$. Please determine the maximum allowable load F .



Deflection table



Notation:

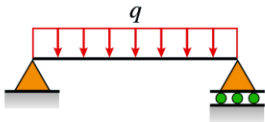
v = deflection in the y direction (positive upward)

$v' = dv/dx$ = slope of the deflection curve

$\delta_B = -v(L)$ = deflection at end B of the beam (positive downward)

$\theta_B = -v'(L)$ = angle of rotation at end B of the beam (positive clockwise)

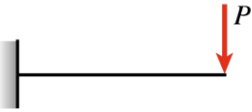
EI = constant



$$v = -\frac{qx}{24EI}(L^3 - 2Lx^2 + x^3)$$

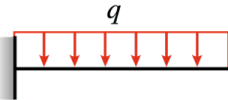
$$v' = -\frac{q}{24EI}(L^3 - 6Lx^2 + 4x^3)$$

$$\delta_C = \delta_{\max} = \frac{5qL^4}{384EI} \quad \theta_A = \theta_B = \frac{qL^3}{24EI}$$



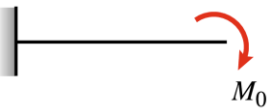
$$v = -\frac{Px^2}{6EI}(3L - x) \quad v' = -\frac{Px}{2EI}(2L - x)$$

$$\delta_B = \frac{PL^3}{3EI} \quad \theta_B = \frac{PL^2}{2EI}$$



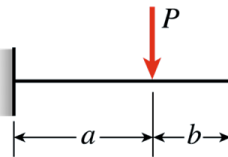
$$v = -\frac{qx^2}{24EI}(6L^2 - 4Lx + x^2) \quad v' = \frac{qx}{6EI}(3L^2 - 3Lx + x^2)$$

$$\delta_B = \frac{qL^4}{8EI} \quad \theta_B = \frac{qL^3}{6EI}$$



$$v = -\frac{M_0x^2}{2EI} \quad v' = -\frac{M_0x}{EI}$$

$$\delta_B = \frac{M_0L^2}{2EI} \quad \theta_B = \frac{M_0L}{EI}$$



$$v = -\frac{Px^2}{6EI}(3a - x) \quad v' = -\frac{Px}{2EI}(2a - x) \quad (0 \leq x \leq a)$$

$$v = -\frac{Pa^2}{6EI}(3x - a) \quad v' = -\frac{Pa^2}{2EI} \quad (a \leq x \leq L)$$

$$\text{At } x = a: \quad v = -\frac{Pa^3}{3EI} \quad v' = -\frac{Pa^2}{2EI}$$

$$\delta_B = \frac{Pa^2}{6EI}(3L - a) \quad \theta_B = \frac{Pa^2}{2EI}$$